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Method for forming semiconductor device with monoclinic crystalline metal oxide capping layer

Abstract

A method includes forming a semiconductor fin protruding over a substrate; forming an isolation structure over the substrate; depositing a first metal oxide layer over the isolation structure; depositing a first oxide layer over the first metal oxide layer; depositing a second metal oxide layer over the first oxide layer, in which the first metal oxide layer and the second metal oxide layer comprise amorphous structures; performing a chemical mechanism polishing (CMP) process to the first metal oxide layer, the first oxide layer, and the second metal oxide layer; after the CMP process is completed, performing an annealing process such that the first metal oxide layer and the second metal oxide layer are transferred from the amorphous structures into crystalline structures; forming a gate structure over the semiconductor fin; and forming source/drain structures over the substrate and on opposite sides of the gate structure.

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Background/Summary

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs. However, since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 illustrates an example of nano-FETs in a three-dimensional view in accordance with some embodiments.
- (3) FIGS. 2A through 19D are cross-sectional views of intermediate stages in the manufacturing of a semiconductor device, in accordance with some embodiments.
- (4) FIG. 20 shows experiment results in accordance with some embodiments of the present disclosure.
- (5) FIG. 21 shows experiment results in accordance with some embodiments of the present disclosure.
- (6) FIG. 22 shows experiment results in accordance with some embodiments of the present disclosure.
- (7) FIGS. 23A, 23B, and 23C are cross-sectional views of intermediate stages in the manufacturing

of a semiconductor device in accordance with some embodiments.

DETAILED DESCRIPTION

(8) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(9) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(10) The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

(11) Embodiments are described below in a particular context, a die comprising nano-FETs. Various embodiments may be applied, however, to dies comprising other types of transistors (e.g., fin field effect transistors (FinFETs), planar transistors, thin film transistors (TFTs), or the like) in lieu of or in combination with the nano-FETs.

(12) FIG. 1 illustrates an example of nano-FETs (e.g., nanowire FETs, nanosheet FETs, or the like) in a three-dimensional view, in accordance with some embodiments. The nano-FETs include nanostructures **55** (e.g., nanosheets, nanowire, or the like) over fins **66** on a substrate **50** (e.g., a semiconductor substrate), wherein the nanostructures **55** act as channel regions for the nano-FETs. The nanostructure **55** may include p-type nanostructures, n-type nanostructures, or a combination thereof. Isolation structures **68** are disposed between adjacent fins **66**, which may protrude above and from between neighboring isolation structures **68**. Additionally, although a bottom portion of the fins **66** are illustrated as being single, continuous materials with the substrate **50**, the bottom portion of the fins **66** and/or the substrate **50** may include a single material or a plurality of materials. In this context, the fins **66** refer to the portion extending between the neighboring isolation structures **68**.

(13) Gate dielectric layers **72** are over the fins **66** and along top surfaces, sidewalls, and bottom surfaces of the nanostructures **55**. Gate electrodes **74** are over the gate dielectric layers **72**. The gate dielectric layer **72** and the gate electrode **74** can be collectively referred to as a gate structure **70**.

(14) Epitaxial source/drain structures **80** are disposed over the fins **66**, and on opposing sides of the gate structure **70**. The gate structure **70** serves as a gate region of the transistor, the epitaxial source/drain structures **80** serve as source/drain regions of the transistor, and the nanostructures **55**

serve as channel region of the transistor.

(15) An isolation structure **90** is disposed over the isolation structures **68** and separates at least two adjacent gate structures **70**. Although the isolation structure **90** is illustrated as being single, continuous material, the isolation structure **90** may include a plurality of layers and/or materials.

(16) FIG. **1** further illustrates reference cross-sections that are used in later figures. Cross-section A-A is parallel to a longitudinal axis of a fin **66** of the nano-FET and in a direction of, for example, a current flow between the epitaxial source/drain structures **80** of the nano-FET. Cross-section B-B is along a longitudinal axis of a gate structure **70** and in a direction, for example, perpendicular to the direction of current flow between the epitaxial source/drain structures **80** of a nano-FET. Cross-section C-C is parallel to cross-section B-B and extends through epitaxial source/drain structures **80** of the nano-FET of the nano-FETs. Subsequent figures refer to these reference cross-sections for clarity.

(17) FIGS. **2A** through **19C** are cross-sectional views of intermediate stages in the manufacturing of a semiconductor device, in accordance with some embodiments. FIGS. **2A-19A** illustrate reference cross-section A-A illustrated in FIG. **1**. FIGS. **2B-19B** illustrate reference cross-section B-B illustrated in FIG. **1**. FIGS. **2C-19C** illustrate reference cross-section C-C illustrated in FIG. **1**.

(18) Reference is made to FIGS. **2A** to **2C**, shown there is a substrate **100**. Generally, the substrate **100** illustrated in FIGS. **2A** to **2C** may include a bulk semiconductor substrate or a silicon-on-insulator (SOI) substrate. An SOI substrate includes an insulator layer below a thin semiconductor layer that is the active layer of the SOI substrate. The semiconductor of the active layer and the bulk semiconductor generally include the crystalline semiconductor material silicon, but may include one or more other semiconductor materials such as germanium, silicon-germanium alloys, compound semiconductors (e.g., GaAs, AlAs, InAs, GaN, AlN, and the like), or their alloys (e.g., Ga.sub.xAl.sub.1-xAs, Ga.sub.xAl.sub.1-xN, In.sub.xGa.sub.1-xAs and the like), oxide semiconductors (e.g., ZnO, SnO.sub.2, TiO.sub.2, Ga.sub.2O.sub.3, and the like) or combinations thereof. The semiconductor materials may be doped or undoped. Other substrates that may be used include multi-layered substrates, gradient substrates, or hybrid orientation substrates.

(19) Further in FIGS. **2A** to **2C**, a plurality of semiconductor layers **103** and semiconductor layers **104** are alternately deposited over the substrate **100**. The semiconductor layers **103** and the semiconductor layers **104** have different materials and/or components, such that the semiconductor layers **103** and the semiconductor layers **104** have different etching rates. In some embodiments, the semiconductor layers **103** are made from SiGe. The germanium percentage (atomic percentage concentration) of the semiconductor layers **103** is in the range between about **10** percent and about **20** percent, while higher or lower germanium percentages may be used. It is appreciated, however, that the values recited throughout the description are examples, and may be changed to different values. For example, the semiconductor layers **103** may be Si.sub.0.8Ge.sub.0.2 or Si.sub.0.9Ge.sub.0.1, in which the proportion between Si and Ge may vary from embodiments, and the disclosure is not limited thereto. The semiconductor layers **104** may be pure silicon layers that are free of germanium. The semiconductor layers **104** may also be substantially pure silicon layers, for example, with a germanium percentage lower than about 1 percent. In some embodiments, the semiconductor layers **103** have a higher germanium atomic percentage concentration than the semiconductor layers **104**. The semiconductor layers **103** and **104** may be formed by chemical vapor deposition (CVD), molecular beam epitaxy (MBE), or other suitable process(es). In some embodiments, the semiconductor layers **103** and **104** are formed by an epitaxy growth process, and thus the semiconductor layers **103** and **104** can also be referred to as epitaxial layers in this content.

(20) Reference is made to FIGS. **3A** to **3C**. The semiconductor layers **103**, **104** and the substrate **100** are patterned to form trenches TR1. As a result of the patterning process, semiconductor strips **102** are formed protruding over the substrate **100**. In some embodiments, the semiconductor layers **103**, **104** and the substrate **100** may be patterned using suitable processes including photolithography and etch processes. In some embodiments, each semiconductor strip **102** and the

overlying semiconductor layers **103** and **104** can be collectively referred to as a semiconductor fin **101**.

(21) Reference is made to FIGS. **4A** to **4C**. Isolation structures **110** are formed over the substrate **100** and laterally surrounding the semiconductor strips **102**. The isolation structures **110** may be shallow trench isolation (STI) structures, suitable isolation structures, combinations of the foregoing, or the like. In some embodiments, the isolation structures **110** may be made of oxide (e.g., silicon oxide) or nitride (e.g., silicon nitride). In some other embodiments, each of the isolation structures **110** may include a dielectric layer and a dielectric liner lining the dielectric layer, in which the dielectric liner and the dielectric layer are made of different materials, for example, the dielectric liner may be silicon nitride, and the dielectric layer may be silicon oxide. In some embodiments, the isolation structures **110** may be formed by, for example, depositing dielectric material(s) over the substrate **100** and overfilling the trenches **TR1**, performing a chemical mechanism polishing (CMP) process to the dielectric material(s), and etching back the dielectric material(s) until top portions of sidewalls of the semiconductor strips **102** are exposed.

(22) Reference is made to FIGS. **5A** to **5C**. Semiconductor layers **125** are formed along sidewalls of the semiconductor layers **103**, **104**, and along sidewalls of the semiconductor strips **102**. In some embodiments, the semiconductor layers **125** may also be referred to as isolation films. The semiconductor layers **125** may include silicon (Si) or silicon germanium (SiGe). In some embodiments, the semiconductor layers **125** may be formed by, for example, depositing a semiconductor layer conformally over the structure of FIGS. **4A** to **4C**, and performing an etching process to remove horizontal portions of the semiconductor layer, such that the vertical portions of the semiconductor layer remain on sidewalls of the semiconductor layers **103**, **104**, and on sidewalls of the semiconductor strips **102**. In some embodiments, the semiconductor layer **125** may be formed by CVD, PVD, ALD, or other suitable deposition processes.

(23) Reference is made to FIGS. **6A** to **6C**. An isolation film **132** is deposited conformally over the structure of FIGS. **5A** to **5C**. In some embodiments, the isolation film **132** may be made of a dielectric material. In particular, the isolation film **132** may be made of a nitride-based material, such as SiCN, SiON, SiCON, or the like. In some embodiments, the isolation film **132** may be deposited by suitable deposition process, such as CVD, PVD, ALD, or other suitable deposition processes.

(24) Reference is made to FIGS. **7A** to **7C**. An isolation film **134** is deposited over the isolation film **132**. In some embodiments, the isolation film **134** may be made of a dielectric material. In particular, the isolation film **134** may be made of an oxide-based material, such as SiO₂, or the like. In some embodiments, the isolation film **134** is made of a different material than the isolation film **132**, so as to provide sufficient etching selectivity between the two layers.

(25) Reference is made to FIGS. **8A** to **8C**. A CMP process is performed to remove excess materials of the isolation films **132** and **134** until the top surfaces of the topmost semiconductor layers **104** are exposed. After the CMP process is completed, top surfaces of the isolation films **132** and **134** are substantially level with the top surface of the topmost semiconductor layer **104**. In some embodiments, the isolation films **132** and **134** can be collectively referred to as an isolation structure **130**.

(26) Reference is made to FIGS. **9A** to **9C**. An etching back process is performed to etch back the isolation films **132** and **134**, such that top surfaces of the isolation films **132** and **134** are lower to positions below the top surfaces of the topmost semiconductor layers **104**. In some embodiments, during the etching back process, the topmost semiconductor layer **104** and the semiconductor layers **125** have higher etching resistance to the etchant of the etching back process, such that the semiconductor layer **104** and the semiconductor layers **125** are substantially intact or negligibly etched after the etching back process. After the etching back process is completed, a plurality of recesses **R1** are formed.

(27) Reference is made to FIG. **9D**, in which FIG. **9D** is an enlarged view of recess **R1** in the

middle of FIGS. **9B** and **9C** (e.g., the recess **R1** between two semiconductor strips **102**). The recess **R1** has a bottom tip having an angle θ in a range from about 60° to about 179° . In some other embodiments, the angle θ of the bottom tip of the recess **R1** may be in a range from about 85° to about 95° , such as 90° . Because isolation films **132** and **134** are made of different materials, the profile of the bottom tip of the recess **R1** is due to the etching selectivity between the isolation films **132** and **134** during the etching back process. In some embodiments, the isolation film **132** has a higher etching resistance to the etching back process than the isolation film **134**. Stated another way, the etching back process has a higher etching rate to the isolation film **134** than to the isolation film **132**. As a result, the isolation film **134** is pulled back to a position even lower than the isolation film **132**. That is, after the etching back process is completed, a bottommost end of the isolation film **134** is lower than a bottommost end of the isolation film **132**. From another view point, after the etching back process is completed, the isolation film **132** has inclined top surfaces. On the other hand, the isolation film **134** has a concave top surface.

(28) FIG. **20** shows experiment results of the etching back process discussed in FIGS. **9A** to **9D** in accordance with some embodiments of the present disclosure. It is shown that when the etching selectivity between the isolation films **132** and **134** increases, the tip angle θ of the recess **R1** decreases accordingly. In some embodiments where the etching selectivity between the isolation films **132** and **134** is around **12**, the angle θ of the bottom tip of the recess **R1** may be in a range from about 85° to about 95° , such as 90° , which will facilitate the formation of air gap (e.g., the air gap **AG** of FIGS. **10B** and **10C**) formed in later step. In some embodiments, the etching selectivity between the isolation films **132** and **134** can be expressed as: $\text{etching selectivity} = \frac{\text{etch rate.sub.isolation films 132}}{\text{etch rate.sub.isolation films 134}}$.

(29) Reference is made to FIGS. **10A** to **10C**. A metal oxide layer **141**, an oxide layer **142**, a metal oxide layer **143**, an oxide layer **144**, and a metal oxide layer **145** are sequentially deposited over the substrate **100**. In greater details, the metal oxide layer **141**, the oxide layer **142**, the metal oxide layer **143** are deposited filling the recesses **R1** (see FIGS. **9B** and **9C**), which will serve as hard mask for protecting the underlying structure during a subsequent etching process.

(30) In some embodiments, the metal oxide layers **141**, **143**, and **145** may be made of high-k dielectric materials, such as ZrO_2 , HfO_2 , or the like. In some embodiments, the oxide layers **142** and **144** may be made of dielectric materials, such as SiO_2 , Al_2O_3 , or the like. In some embodiments, the oxide layers **142** and **144** are free of metal element of the metal oxide layers **141**, **143**, and **145**.

(31) In some embodiments, the metal oxide layers **141**, **143**, and **145** each has higher dielectric constant than those of the oxide layers **142** and **144**. For example, the dielectric constants of the metal oxide layers **141**, **143**, and **145** are in a range from about 10 to about 30, while the dielectric constants of the oxide layers **142** and **144** are in a range from about 2 to about 10.

(32) In some embodiments, the metal oxide layers **141**, **143**, and **145** each has a thickness in a range from about 10 nm to about 40 nm. In some embodiments, the metal oxide layers **141**, **143**, and **145** are amorphous, and can be referred to as amorphous metal oxide layers throughout the content. The amorphous metal oxide layers **141**, **143**, and **145** will be crystallized in following steps, which is beneficial for lowering the effective k-value of the hard mask structure. If the thickness of the metal oxide layers **141**, **143**, and **145** is too small (e.g., much lower than 10 nm), the metal oxide layers **141** and **143** may not be thick enough to act as a protective layer. If the thickness of the metal oxide layers **141**, **143**, and **145** is too large (e.g., much greater than 40 nm), the metal oxide layers **141**, **143**, and **145** may be naturally crystallized into an unwanted phase during the deposition, which will deteriorate the device performance.

(33) In some embodiments, the oxide layers **142** and **144** each has a thickness in a range from about 1 nm to about 1.5 nm. That is, the oxide layers **142** and **144** may be thinner than the metal oxide layers **141**, **143**, and **145**. In some embodiments, the oxide layer **142** is an interposed layer between the metal oxide layers **141** and **143**, and oxide layer **144** is an interposed layer between the

metal oxide layers **143** and **145**. Such configuration is beneficial for preventing the formation of unwanted crystallized metal oxide layer. For example, if the oxide layers **142** and **144** are omitted, a thick metal oxide layer will be deposited over the substrate **100** and filling the recesses **R1**, and will be naturally crystallized into an unwanted phase during the deposition. Accordingly, one advantage of forming the oxide layers **142** and **144** is that the oxide layers **142** and **144** can act as interposers which allow several thin amorphous metal oxide layers (e.g., the metal oxide layers **141**, **143**, and **145**) to be formed. Another advantage of forming the oxide layers **142** and **144** is that the oxide layers **142** and **144** can lower the k-value of the resulting structure, which will improve the device performance.

(34) During the deposition of the metal oxide layer **143**, an air gap AG is formed in the metal oxide layer **143** within the recesses **R1**. This is because the bottom tip of the recesses **R1** as discussed in FIGS. **9A** to **9D**, which is resulted from the etching selectivity between the isolation films **132** and **134** during the etching back process discussed in FIGS. **9A** to **9D**. For example, when the metal oxide layer **141** and the oxide layer **142** are deposited in the recesses **R1**, the middle region of each recess **R1** is unoccupied. Because bottom tip of each recess **R1** will increase aspect ratio of the unoccupied region of each recess **R1**, the metal oxide layer **143** will be deposited filling into a high aspect ratio recess, and thus the air gap AG may be easily formed. One advantage of forming the air gap AG is that the air gap AG can lower the k-value of the resulting structure, which will improve the device performance.

(35) Reference is made to FIGS. **11A** to **11C**. A CMP process is performed to remove excess materials of the metal oxide layer **141**, the oxide layer **142**, the metal oxide layer **143**, the oxide layer **144**, and the metal oxide layer **145** until the top surfaces of the topmost semiconductor layers **104** are exposed. As a result of the CMP process, hard masks **140** are formed, in which each hard mask **140** includes remaining portions of the metal oxide layer **141**, the oxide layer **142**, and the metal oxide layer **143**. In some embodiments, the oxide layer **144** and the metal oxide layer **145** may be completely removed from the substrate **100** during the CMP process, such that the hard masks **140** are free of materials of the oxide layer **144** and the metal oxide layer **145**. In some embodiments, the hard masks **140** can be interchangeably referred to as capping layers throughout the content.

(36) Reference is made to FIGS. **12A** to **12C**. An annealing process AN is performed to crystallize the metal oxide layers **141** and **143** of the hard masks **140**. As mentioned above, the metal oxide layers **141** and **143** are deposited having amorphous structure, and the annealing process AN is performed to change the amorphous metal oxide layers **141** and **143** into desired crystalline phase. In some embodiments, the crystallized metal oxide layers **141** and **143** may include monoclinic phase, tetragonal phase, cubic phase, and orthorhombic phase. However, the dominated phase of the crystallized metal oxide layers **141** and **143** is monoclinic phase. That is, the crystallized metal oxide layers **141** and **143** may include monoclinic phase, tetragonal phase, cubic phase, and orthorhombic phase, while the monoclinic phase has a highest percentage among all crystalline phase. In some embodiments, the percentage of the monoclinic phase of each of the metal oxide layers **141** and **143** is in a range from about 40% to about 100%. For example, in some embodiments, the percentage of the monoclinic phase of each of the metal oxide layers **141** and **143** is about 40%, while other 60% of each of the metal oxide layers **141** and **143** may include a combination of tetragonal phase, cubic phase, and orthorhombic phase.

(37) FIG. **21** shows experiment results in accordance with some embodiments of the present disclosure. It can be seen that, on the left side of FIG. **21**, when the crystallinity of the metal oxide layer **141** (or **143**) increases from 0% (e.g., amorphous) to 100%, the effective k-value will decrease accordingly. Furthermore, on the right side of FIG. **21**, when the percentage of the monoclinic phase of the metal oxide layer **141** (or **143**) increases from 0% to 100%, the effective k-value will decrease accordingly. The results show that the increased crystallinity and the increased percentage of the monoclinic phase are beneficial for lowering the effective k-value of the hard

masks **140**.

(38) Referring back to FIGS. **12A** to **12C**, in some embodiments, the annealing process AN can be a rapid thermal annealing (RTA) process. The annealing process AN may be performed under a temperature in a range from about 700° C. to about 1300° C. In some embodiments, the annealing process AN may be performed under a temperature greater than about 900° C., such as about 900° C. to about 1300° C. The duration of the annealing process AN may be in a range from about 0.1 ms to about 300 s. In some embodiments, the duration of the annealing process AN may be lower than about 30 s, such as about 0.1 ms to about 30 s. In some embodiments, the annealing process AN may be performed in about 0.05% to about 100% oxygen (O.sub.2) gas environment under a pressure in a range from about 5 torr to about 100 torr. If the annealing process AN is performed under a condition beyond the above mentioned ranges, the dominated phase of the crystallized metal oxide layers **141** and **143** may not be monoclinic phase, which will resulting in an unsatisfying device performance.

(39) Reference is made to FIGS. **13A** to **13C**. An etching back process is performed to the semiconductor layers **103**, **104**, and **125**, so as to lower top surfaces of the semiconductor layers **103**, **104**, and **125** to positions lower than the top surfaces of the hard masks **140**. After the etching back process is completed, recesses R2 are formed over the semiconductor layers **103**, **104**, and **125**, and between two adjacent hard masks **140**. In some embodiments, the recesses R2 expose at least portions of the isolation film **132** of the isolation structure **130**.

(40) Reference is made to FIGS. **14A** to **14C**. Dummy gate structures **150** are formed over the semiconductor layers **103**, **104**, and gate spacers **155** are formed on opposite sidewalls of the dummy gate structure **330**. In some embodiments, the dummy gate structure **150** may be formed by, for example, depositing a gate dielectric layer **152** and a gate electrode **154** over the semiconductor layers **103**, **104**, performing a CMP process until the top surface of the dielectric layer **176** is exposed, and subsequently performing a patterning process. The remaining portions of the gate dielectric layer **152** and the gate electrode **154** can be collectively referred to as dummy gate structures **150**. In some embodiments, the gate spacers **155** may be formed by, for example, depositing a spacer material blanket over the dummy gate structures **150**, and subsequently performing an etching process to remove horizontal portions of the spacer material, such that vertical portions of the spacer material remains on sidewalls of the dummy gate structures **150**. In the cross-sectional view of FIG. **14B**, each of the dummy gate structures **150** is formed between adjacent hard masks.

(41) In some embodiments, the gate dielectric layer **152** may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. The gate dielectric layer **152** may be formed by a suitable process, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or any suitable process. The gate electrode **154** may include polycrystalline-silicon (poly-Si) or poly-crystalline silicon-germanium (poly-SiGe). Further, the gate electrode **154** may be doped poly-silicon with uniform or non-uniform doping. The gate electrode **154** may be formed by a suitable process, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or any suitable process.

(42) Reference is made to FIGS. **15A** to **15C**. The semiconductor layers **103**, **104** are recessed to form recesses R3, so as to expose the top surfaces of the semiconductor strips **102**. In greater details, the semiconductor layers **103**, **104** are recessed by using the hard masks **140**, the dummy gate structures **150**, and the gate spacers **155** as etch masks. Afterward, the semiconductor layers **103** are etched, and a plurality of inner spacers **158** are formed vertically between the semiconductor layers **104**. In some embodiments, a first etching process is performed to remove portions of the semiconductor layers **103**, **104** not covered by the dummy gate structures **150** to form the recesses R3. Then, a second etch process is performed to laterally shorten the semiconductor layers **103** through the recesses R3, so as to form spaces between two adjacent

semiconductor layers **104**. Next, inner spacers **158** are formed in the spaces between two adjacent semiconductor layers **104** by a suitable deposition process. For example, the inner spacers **158** may be formed by depositing a spacer material blanket over the substrate **100** and subsequently performing a patterning process to remove portions of the spacer material, such that the remaining portions of the spacer material are left in the spaces between two adjacent semiconductor layers **104**.

(43) Reference is made to FIGS. **16A** to **16C**. Epitaxy source/drain structures **160** are formed over the semiconductor strips **102** and on opposite sides of the dummy gate structures **150**. In some embodiments, the epitaxy source/drain structures may be formed by selective epitaxial growth (SEG). In various embodiments, the epitaxy structures **160** may include Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, or other suitable material. In some embodiments, the epitaxy structures **160** may be doped with p-type dopants or n-type dopants.

(44) Reference is made to FIGS. **17A** to **17C**. An interlayer dielectric (ILD) layer **170** is formed over the epitaxy structures **160** and laterally surrounding the dummy gate structures **150**. In some embodiments, the ILD layer **170** may be formed by, for example, depositing an ILD material layer over the substrate **100** and subsequently performing a CMP process to remove the excess ILD material layer until the top surfaces of the dummy gate structures **150** are exposed. In some embodiments, the ILD layer **170** may include silicon oxide, silicon nitride, silicon oxynitride, tetraethoxysilane (TEOS), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), low-k dielectric material, and/or other suitable dielectric materials. Examples of low-k dielectric materials include, but are not limited to, fluorinated silica glass (FSG), carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), or polyimide.

(45) Reference is made to FIGS. **18A** to **18C**. The dummy gate structures **150**, the semiconductor layers **103**, and the semiconductor layer **125** are removed to form gate trenches TR2. In FIG. **18A**, each gate trench TR2 is between adjacent gate spacers **155**. In FIG. **18B**, each gate trench TR2 at least exposes sidewalls of the isolation structure **130** and sidewalls of the hard masks **140**. In some embodiments, the dummy gate structures **150**, the semiconductor layers **103**, and the semiconductor layer **125** may be removed by a suitable process, such as wet etch, dry etch, or combinations thereof.

(46) Reference is made to FIGS. **19A** to **19C**. Metal gate structures **180** are formed in the gate trenches TR2. In some embodiments, the gate structures **180** include an interfacial layer **182**, a gate dielectric layer **184** over the interfacial layer **182**, and a gate conductive layer **186** over the gate dielectric layer **184**. In some embodiments, the gate structures **180** may be formed by, for example, forming an interfacial material selectively on the exposed semiconductor layers **104**, depositing a gate dielectric material over the interfacial material, depositing a gate conductive material over the gate dielectric material, and subsequently performing a CMP process until the top surface of the ILD layer **170** is exposed.

(47) As shown in FIG. **19B**, the gate dielectric layers **184** of the gate structures **180** extend along the sidewalls of the isolation structures **130** and the sidewalls of the hard masks **140**. In greater details, the gate dielectric layers **184** of the gate structures **180** are in contact with the isolation film **132** of the isolation structures **130**, and are in contact with the metal oxide layer **141** of the hard masks **140**.

(48) In some embodiments, the interfacial layer **182** may be made of oxide, such as silicon oxide (SiO₂). In some embodiments, the interfacial layer **182** may be formed by an oxidation process, such as a thermal oxidation process.

(49) In some embodiments, the gate dielectric layers **184** may be made of high-k dielectric materials, such as metal oxides, transition metal-oxides, or the like. Examples of the high-k dielectric material include, but are not limited to, hafnium oxide (HfO₂), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), zirconium oxide, titanium oxide, aluminum oxide, hafnium dioxide-alumina

(HfO.sub.2—Al.sub.2O.sub.3) alloy, or other applicable dielectric materials. In some embodiments, the gate dielectric layer **184** may include oxide layers. The gate dielectric layer **184** may be formed by PVD, CVD, ALD, or other suitable deposition processes.

(50) In some embodiments, the gate conductive layer **186** may include a work function metal layer and a filling metal. The work function metal layer may be an n-type or p-type work function layer. Exemplary p-type work function metals include TiN, TaN, Ru, Mo, Al, WN, ZrSi.sub.2, MoSi.sub.2, TaSi.sub.2, NiSi.sub.2, WN, other suitable p-type work function materials, or combinations thereof. Exemplary n-type work function metals include Ti, Ag, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, other suitable n-type work function materials, or combinations thereof. The work function layer may include a plurality of layers. The filling metal may include tungsten (W), aluminum (Al), copper (Cu), or another suitable conductive material(s). The gate conductive layer **186** may be formed by PVD, CVD, ALD, or other suitable deposition processes.

(51) Reference is made to FIG. **19D**, in which FIG. **19D** is an enlarged view of FIG. **19B**. After the metal gate structures **180** are formed, the resulting structure may include at least two gate structures **180**, and may include an isolation structure **130** and a hard mask **140** that laterally separates the metal gate structures **180**.

(52) In greater details, the isolation structure **130** includes an isolation film **132** and an isolation film **134** over the isolation film **132**, in which the isolation film **132** has two vertical portions lining opposite sidewalls of the isolation film **134**. The vertical portions of the isolation film **132** has a lateral thickness a, and the isolation film **134** has a lateral thickness b. In some embodiments, the thickness a is greater than the thickness b. In other embodiments, the thickness a is lower than the thickness b. In yet other embodiments, the thickness a is substantially equal the thickness b.

(53) The hard mask **140** includes a metal oxide layer **141**, an oxide layer **142** over the metal oxide layer **141**, and metal oxide layer **143** over the oxide layer **142**. The hard mask **140** has a height H and a width W. In some embodiments, the height H of the hard mask **140** is in a range from about 10 nm to about 30 nm. In some embodiments, the width W of the hard mask **140** is in a range from about 10 nm to about 2000 nm. In one embodiment, the height H of the hard mask **140** is about 25 nm, and the width W of the hard mask **140** is about 18 nm. Here, the “height” of the hard mask **140** is the vertical length of the sidewall of the hard mask **140** that is in contact with the gate structure **180**.

(54) In some embodiments, the metal oxide layers **141** and **143** are thicker than the oxide layer **142**. The oxide layer **142** has a thickness d. In some embodiments, the thickness d of the oxide layer **142** is in a range from about 0 nm to about 1.5 nm. The metal oxide layer **143** includes an air gap AG. In some embodiments, in the cross-sectional view of FIG. **19D**, the ratio of the area of the air gap AG to the area of the hard mask HM is in a range from about 0% to about 10%. In one embodiment, the thickness d of the oxide layer **142** is about 0.5 nm, and the area ratio of the air gap AG is about 2%. In some embodiments, the oxide layer **142** may be omitted. In some embodiments, the air gap AG may not be formed in the metal oxide layer **143**.

(55) In some embodiments, the hard mask **140** has a bottom tip **140T**. In some embodiments, the bottom tip **140T** has a rounded surface. In some embodiments, the bottom tip **140T** of the hard mask **140** is formed by two inclined surfaces, in which the two inclined surfaces form an angle θ there between. In some embodiments, the angle θ is in a range from about 60° to about 179°. In some other embodiments, the angle θ of the bottom tip of the recess R1 may be in a range from about 85° to about 95°, such as 90°. In some embodiments, a bottommost end of the metal oxide layer **141** of the hard mask **140** is lower than topmost ends of the isolation films **132** and **134** of the isolation structure **130**.

(56) FIG. **22** shows experiment results in accordance with some embodiments of the present disclosure. It can be seen that, on the left side of FIG. **22**, when the area ratio of the air gap AG increases, the effective k-value will decrease accordingly. Furthermore, on the right side of FIG. **22**, when the thickness d of the oxide layer **142** increases, the effective k-value will decrease

accordingly. The results show that the formations of oxide layer **142** and the air gap AG are beneficial for lowering the effective k-value of the hard mask **140**.

(57) FIGS. **23A**, **23B**, and **23C** are cross-sectional views of intermediate stages in the manufacturing of a semiconductor device in accordance with some embodiments. The cross-sectional views of FIGS. **23A**, **23B**, and **23C** are the same as the cross-sectional views of FIGS. **19A**, **19B**, and **19C**. Furthermore, some elements of FIGS. **23A**, **23B**, and **23C** are the same as those described in FIGS. **19A**, **19B**, and **19C**, such elements are labeled the same, and relevant details will not be repeated for brevity.

(58) FIGS. **23A**, **23B**, and **23C** are different from FIGS. **19A**, **19B**, and **19C**, in that the semiconductor device shown in FIGS. **23A**, **23B**, and **23C** is a fin-type field effect transistor. For example, semiconductor fins **202** are disposed over the substrate **100**. The gate structures **180** may cross the corresponding semiconductor fins **202**. In some embodiments, the semiconductor fins **202** may be made of silicon, germanium, or silicon-germanium alloy ($\text{Si}_{1-x}\text{Ge}_x$, $0 \leq x \leq 1$). In some embodiments, the semiconductor fins **202** can be formed by patterning the substrate **100**. Other structures shown in FIGS. **23A**, **23B**, and **23C** may be formed by using similar methods as described in FIGS. **2A** to **19C**, and thus relevant details will not be repeated.

(59) Based on the above discussions, it can be seen that the present disclosure offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. One advantage of the embodiments is that by forming an amorphous metal oxide hard mask, followed by an annealing process to change the amorphous metal oxide hard mask into crystallized metal oxide hard mask, which can improve the etching resistance for following etching process. Another advantage of the embodiments is that by controlling the conditions of the annealing process, the major phase of the crystallized metal oxide hard mask will be monoclinic phase, which can lower the effective k-value of the hard mask. Yet another advantage of the embodiments is that by forming an oxide layer and air gap in the metal oxide hard mask, which can further lower the effective k-value of the hard mask.

(60) In some embodiments of the present disclosure, a method includes forming a semiconductor fin protruding over a substrate; forming an isolation structure over the substrate; depositing a first metal oxide layer over the isolation structure; depositing a first oxide layer over the first metal oxide layer; depositing a second metal oxide layer over the first oxide layer, in which the first metal oxide layer and the second metal oxide layer comprise amorphous structures; performing a chemical mechanical polishing (CMP) process to the first metal oxide layer, the first oxide layer, and the second metal oxide layer; after the CMP process is completed, performing an annealing process such that the first metal oxide layer and the second metal oxide layer are transferred from the amorphous structures into crystalline structures; forming a gate structure over the semiconductor fin; and forming source/drain structures over the substrate and on opposite sides of the gate structure.

(61) In some embodiments, the method further includes depositing a second oxide layer over the second metal oxide layer; and depositing a third metal oxide layer over the second oxide layer, in which the CMP process remove an entirety of the second oxide layer and the third metal oxide layer from the substrate.

(62) In some embodiments, forming the isolation structure includes depositing a first isolation film over the substrate; and depositing a second isolation film over the first isolation film, in which the second isolation film is made of a different material than the first isolation film.

(63) In some embodiments, the method further includes performing an etching back process to the isolation structure prior to depositing the first metal oxide layer, in which the first isolation film has a higher etching resistance to the etching back process than that of the second isolation film, such that a top surface of the second isolation film is lower than a top surface of the first isolation film after the etching back process is completed.

(64) In some embodiments, an air gap is formed in the second metal oxide layer during depositing the second metal oxide layer.

(65) In some embodiments, the first oxide layer has a lower dielectric constant than those of the first metal oxide layer and the second metal oxide layer.

(66) In some embodiments, the method further includes etching back the isolation structure such that the isolation structure has a concave top surface.

(67) In some embodiments, the method further includes forming a shallow trench isolation (STI) structure over the substrate prior to forming the isolation structure, in which isolation structure is in contact with a top surface of the STI structure.

(68) In some embodiments, the method further includes forming a semiconductor layer along a sidewall of the semiconductor fin prior to depositing the first metal oxide layer, in which the first metal oxide layer is in contact with the semiconductor layer.

(69) In some embodiments, the annealing process is performed under a temperature in a range from about 700° C. to about 1300° C., the annealing process is performed for a duration in a range from 0.1 ms to about 300 s, and the annealing process is performed under a pressure in a range from about 5 torr to about 1000 torr.

(70) In some embodiments of the present disclosure, a method includes forming first and second semiconductor layers alternately arranged over a substrate; depositing a third semiconductor layer along sidewalls of the first and second semiconductor layers; depositing a first isolation film over the substrate and along a sidewall of the third semiconductor layer; depositing a second isolation film over the first isolation film; performing an etching back process to the first and second isolation films, in which the first isolation film has a higher etching resistance to the etching back process than that of the second isolation film, such that a top surface of the first isolation film is higher than a top surface of the second isolation film after the etching back process is completed; depositing a first metal oxide layer over the first and second isolation films; depositing a first oxide layer over the first metal oxide layer; depositing a second metal oxide layer over the first oxide layer, in which an air gap is formed in the second metal oxide layer; removing the first semiconductor layers; forming a gate structure wrapping the second semiconductor layers; and forming source/drain structures over the substrate and on opposite sides of the gate structure.

(71) In some embodiments, the method further includes performing a chemical mechanism polishing (CMP) process to the first metal oxide layer, the first oxide layer, and the second metal oxide layer; and after the CMP process is completed, performing an annealing process such that the first metal oxide layer and the second metal oxide layer are transferred from amorphous structures into crystalline structures.

(72) In some embodiments, the method further includes depositing a second oxide layer over the second metal oxide layer; and depositing a third metal oxide layer over the second oxide layer.

(73) In some embodiments, removing the first semiconductor layers further removes the third semiconductor layer.

(74) In some embodiments, the first oxide layer is thinner than the first metal oxide layer and the second metal oxide layer.

(75) In some embodiments of the present disclosure, a semiconductor device includes a substrate, nanostructures, a gate structure, source/drain structures, an isolation structure, and a capping layer. The nanostructures are alternately arranged over the substrate. The gate structure is over the substrate and wraps each of the nanostructures. The source/drain structures are over the substrate and on opposite sides of the gate structure. The isolation structure is over the substrate and along a sidewall of the gate structure. The capping layer is over the isolation structure and along the sidewall of the gate structure, in which the capping layer includes a first metal oxide layer, an oxide layer, and a second metal oxide layer, in which crystalline phases of the first metal oxide layer include a monoclinic phase, and the monoclinic phase of the crystalline phases of the first metal oxide layer has a highest percentage among the crystalline phases of the first metal oxide layer.

(76) In some embodiments, the crystalline phases of the first metal oxide layer further include a tetragonal phase, a cubic phase, and an orthorhombic phase.

(77) In some embodiments, the second metal oxide layer includes an air gap.

(78) In some embodiments, crystalline phases of the second metal oxide layer include a monoclinic phase, and the monoclinic phase of the crystalline phases of the second metal oxide layer has a highest percentage among the crystalline phases of the second metal oxide layer.

(79) In some embodiments, first metal oxide layer and the second metal oxide layer are made of a same material that has a higher dielectric constant than that of the oxide layer.

(80) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a semiconductor fin protruding over a substrate; forming an isolation structure over the substrate; depositing a first metal oxide layer over the isolation structure; depositing a first oxide layer over the first metal oxide layer; depositing a second metal oxide layer over the first oxide layer, wherein the first metal oxide layer and the second metal oxide layer comprise amorphous structures; performing a chemical mechanism polishing (CMP) process to the first metal oxide layer, the first oxide layer, and the second metal oxide layer; after the CMP process is completed, performing an annealing process such that the first metal oxide layer and the second metal oxide layer are transferred from the amorphous structures into crystalline structures; forming a gate structure over the semiconductor fin; and forming source/drain structures over the substrate and on opposite sides of the gate structure.
2. The method of claim 1, further comprising: depositing a second oxide layer over the second metal oxide layer; and depositing a third metal oxide layer over the second oxide layer, wherein the CMP process remove an entirety of the second oxide layer and the third metal oxide layer from the substrate.
3. The method of claim 1, wherein forming the isolation structure comprises: depositing a first isolation film over the substrate; and depositing a second isolation film over the first isolation film, wherein the second isolation film is made of a different material than the first isolation film.
4. The method of claim 3, further comprising performing an etching back process to the isolation structure prior to depositing the first metal oxide layer, wherein the first isolation film has a higher etching resistance to the etching back process than that of the second isolation film, such that a top surface of the second isolation film is lower than a top surface of the first isolation film after the etching back process is completed.
5. The method of claim 1, wherein an air gap is formed in the second metal oxide layer during depositing the second metal oxide layer.
6. The method of claim 1, wherein the first oxide layer has a lower dielectric constant than those of the first metal oxide layer and the second metal oxide layer.
7. The method of claim 1, further comprising etching back the isolation structure such that the isolation structure has a concave top surface.
8. The method of claim 1, further comprising forming a shallow trench isolation (STI) structure over the substrate prior to forming the isolation structure, wherein isolation structure is in contact with a top surface of the STI structure.

9. The method of claim 1, further comprising forming a semiconductor layer along a sidewall of the semiconductor fin prior to depositing the first metal oxide layer, wherein the first metal oxide layer is in contact with the semiconductor layer.
10. The method of claim 1, wherein the annealing process is performed under a temperature in a range from about 700° C. to about 1300° C., the annealing process is performed for a duration in a range from 0.1 ms to about 300s, and the annealing process is performed under a pressure in a range from about 5 torr to about 1000 torr.
11. A method, comprising: forming first and second semiconductor layers alternately arranged over a substrate; depositing a third semiconductor layer along sidewalls of the first and second semiconductor layers; depositing a first isolation film over the substrate and along a sidewall of the third semiconductor layer; depositing a second isolation film over the first isolation film; performing an etching back process to the first and second isolation films, wherein the first isolation film has a higher etching resistance to the etching back process than that of the second isolation film, such that a top surface of the first isolation film is higher than a top surface of the second isolation film after the etching back process is completed; depositing a first metal oxide layer over the first and second isolation films; depositing a first oxide layer over the first metal oxide layer; depositing a second metal oxide layer over the first oxide layer, wherein an air gap is formed in the second metal oxide layer; removing the first semiconductor layers; forming a gate structure wrapping the second semiconductor layers; and forming source/drain structures over the substrate and on opposite sides of the gate structure.
12. The method of claim 11, further comprising: performing a chemical mechanism polishing (CMP) process to the first metal oxide layer, the first oxide layer, and the second metal oxide layer; and after the CMP process is completed, performing an annealing process such that the first metal oxide layer and the second metal oxide layer are transferred from amorphous structures into crystalline structures.
13. The method of claim 11, further comprising: depositing a second oxide layer over the second metal oxide layer; and depositing a third metal oxide layer over the second oxide layer.
14. The method of claim 11, wherein removing the first semiconductor layers further removes the third semiconductor layer.
15. The method of claim 11, wherein the first oxide layer is thinner than the first metal oxide layer and the second metal oxide layer.
16. A method, comprising: forming a semiconductor layer over a substrate; forming source/drain structures over the substrate and on opposite sides of the semiconductor layer; forming an isolation structure over the substrate; and forming a capping layer over the isolation structure, wherein the capping layer comprises: a first metal oxide layer, wherein crystalline phases of the first metal oxide layer comprise a monoclinic phase, and the monoclinic phase of the crystalline phases of the first metal oxide layer has a highest percentage among the crystalline phases of the first metal oxide layer; an oxide layer; and a second metal oxide layer; and forming a gate structure wrapping around the semiconductor layer and along the isolation structure and the capping layer.
17. The method of claim 16, wherein the crystalline phases of the first metal oxide layer further comprise a tetragonal phase, a cubic phase, and an orthorhombic phase.
18. The method of claim 16, wherein the second metal oxide layer comprises an air gap.
19. The method of claim 16, wherein the first metal oxide layer and the second metal oxide layer are made of a same material that has a higher dielectric constant than that of the oxide layer.
20. The method of claim 16, further comprising performing a chemical mechanism polishing (CMP) process to the first metal oxide layer, the oxide layer, and the second metal oxide layer prior to forming the gate structure.
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